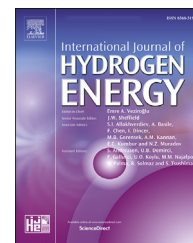


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# Methanol production by CO<sub>2</sub> hydrogenation: Analysis and simulation of reactor performance

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## ABSTRACT

Methanol is a very valuable chemical with a variety of uses, either as a fuel or as building block for the synthesis of other chemicals. In the last years, interest was growing in the production of methanol from CO<sub>2</sub>, based on the so called “Power-to-Fuel” concept. In this research, an equilibrium analysis of a methanol reactor with pure CO<sub>2</sub> and H<sub>2</sub> in the feeding stream was developed. Three novel reactor configurations at equilibrium conditions were considered: once-through reactor, reactor with recycle of unconverted gases after separation of methanol and water by condensation; reactor equipped with membrane permeable to water. An additional important feature of this work was the development of a methodology that assists in comparison of different process schemes by simulation of two different methanol plants configurations in ChemCad®. An adiabatic kinetic reactor with recycle of unconverted gases was considered and simulated in Aspen Plus®, while the performance of a methanol reactor with heat exchange at the pipe wall was simulated in MATLAB. Results show that at equilibrium conditions a reactor with the recycle of unconverted gases ensures the highest CO<sub>2</sub> conversion: 69% at 473 K and 55 bar. In addition, the use of pure CO<sub>2</sub> and H<sub>2</sub> in the feeding stream allows an overall reaction enthalpy change lower than that obtained by the use of syngas in the feed. The kinetic simulation of the methanol reactor in MATLAB showed that axial dispersion phenomena are negligible and the effect of the global heat exchange coefficient on reactor performance is less important than the effect of isothermal heat exchange fluid temperature.

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## Introduction

Today CO<sub>2</sub> hydrogenation reactions are important routes to produce valuable chemicals, such as methanol, ethanol, methane, formic acid, dimethyl ether, syngas. The goal is to valorize CO<sub>2</sub>, reduce CO<sub>2</sub> emissions and satisfy the demand of the above chemicals. For this kind of reactions, the efficiency

of H<sub>2</sub> utilization in the molecular structure of product species is equal to 67%, 50%, 50%, 100%, 50% and 100%, respectively for methanol, ethanol, methane, formic acid, dimethyl ether and syngas. In addition, the energetic efficiency can be calculated: 25.5 kcal/H atoms for methanol, 24.5 kcal/H atoms for ethanol, 24 kcal/H atoms for methane, 40 kcal/H atoms for formic acid, 28.5 kcal/H atoms for dimethyl ether and 24 kcal/H atoms for syngas. As a result, methanol has a high efficiency in H<sub>2</sub>

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utilization and an acceptable value of energetic efficiency, considering that the maximum value is 40 kcal/H atoms for formic acid, where no losses of  $H_2$  are present to form  $H_2O$  in the respective stoichiometric equation.

Ergo, methanol is one of the most valuable chemicals obtainable by  $CO_2$  hydrogenation through the “Power-to-Fuel” concept, with a variety of uses [1]. Methanol is generally used as a solvent and/or as a raw material for the production of different chemicals such as: formaldehyde, acetic acid, methyl methacrylate, dimethyl terephthalate, methylamines, dimethyl ether, methyl-tert-butylether and chloromethanes [2,3]. Also, light olefins (ethylene and propylene) can be obtained with a methanol-to-olefin process [4,5]. In addition, methanol can be used as an alternative fuel in the transportation sector [6,7] and in fuel cell applications [8].

Global methanol production has increased from 32 to 62 million metric tons annually between 2006 and 2012, and for 2016 the demand has already risen to 94 million metric tons [9]. It is necessary to underline that today methanol is preferentially produced by reforming natural gas, i.e. by syngas as feeding stream, and not yet by  $CO_2$  hydrogenation [10,11]. More than 80% of the methanol produced worldwide is obtained from natural gas [12]. However, the conversion of  $CO_2$  into methanol via direct hydrogenation is less exothermic than the conversion of syngas into methanol and leads to a reduced amount of by-products and emissions. Both routes are affected by thermodynamic and kinetic constraints [5]. Pilot and demonstration plants based on  $CO_2$  hydrogenation were realized in Japan, in Osaka in particular, (with a capacity of 100 ton/year, built by Mitsui Chemicals), and Iceland (with a capacity of 4000 ton/year, realized by Carbon Recycling International and operating since 2011) using renewable hydrogen [13,14]. Such technical and financial efforts, although developed under peculiar circumstances characterizing the general context, show that a concrete economic interest accompanies research developments in this field. These can be considered “green processes”, especially when also  $H_2$  is obtained by green and renewable sources as electrolysis of water [15,16], coke oven gas (COG) [17] and chlor-alkali plants [18].  $CO_2$  instead can be captured from power plants, petrochemicals, cement industry, paper mill, air and steel mills, etc. [19].

Atsonios et al. [14] found that the cost linked to hydrogen availability largely influences the total cost of a process producing methanol via  $CO_2$  hydrogenation. In the process studied there, carbon dioxide was captured from a power plant or other intensive carbon emission industry (cement plant and steel industry), through MEA absorption. Hydrogen was produced by an alkaline electrolyzer considering different electric sources as grid, thermal power plant and wind energy.

Bellotti et al. [20] studied a methanol plant where methanol was produced via  $CO_2$  hydrogenation: carbon dioxide was captured from a power plant using amine solution while hydrogen was obtained by a PEM electrolyzer. An economic feasibility study was carried out by means of W-ECOMP (Web-based Economic Cogeneration Modular Program) software considering three different sizes of the plant (4000, 10000, 50000 MeOH ton/year) and a sensitivity analysis was performed varying the methanol selling price, the oxygen selling option for industrial applications and the capital cost of PEM

electrolyzers. This analysis showed that the PEM electrolyzer affects the total capital investment cost significantly (>75%), while a larger plant leads to even higher capital investment.

Perez-Fortes et al. [21] suggested both a simple and an integrated carbon capture and utilization (CCU) system where the captured  $CO_2$  was used to produce methanol via hydrogenation. Carbon dioxide was obtained from a pulverized coal power plant, while hydrogen by water electrolysis. A simulation in ChemCad® of the two processes was carried out. Results showed that in order to make the CCU plant financially attractive, the price of methanol should increase by a factor of almost 2,  $H_2$  costs should decrease almost 2.5 times, or  $CO_2$  should have a value of around 222 €/ton.

Kiss et al. [18] developed a new process for methanol production via  $CO_2$  hydrogenation, using wet  $H_2$  (saturated with water) available as by-product in chlor-alkali production. The key feature of this novel process was the use of a stripping unit where the wet hydrogen flows in counter-current with the condensed mixture of methanol–water resulting from a flash separation after reaction. This process, simulated in Aspen Plus®, produced 100 kton/year of methanol with a yield of 99.83%: 1 ton of chlorine leads to 28 kg of hydrogen that can be further converted into 149 kg of methanol, using 205 kg  $CO_2$ . A sensitivity analysis was carried out to evaluate the effect of temperature, pressure and gas hourly space velocity (GHSV) on methanol yield. Results showed that a higher methanol yield is obtained at lower temperature, higher pressure and lower GHSV, as expected.

In the work of Harp et al. [17], hydrogen was obtained by COG (coke oven gas): the integration of methanol plant and steelworks was proposed. In particular, hydrogen was recovered by pressure swing adsorption (PSA), with a flow rate corresponding to 133 MWth while carbon dioxide was captured from power plant flue gas. The first plant integrating methanol production and steelworks was constructed in China, where  $H_2$  is recovered with a purity equal to 99% [22].

Van-Dal and Bouallou [23] carried out a simulation of methanol plant in Aspen Plus® using the kinetics of Van den Bussche and Froment [24]. Results showed that 1.6 ton of  $CO_2$  are consumed per ton of produced methanol. However, the results are presented without economic assessment and without comparison with other kinetic models.

Rivarolo et al. [25] performed the thermo-economic analysis of a methanol plant for which  $CO_2$  is obtained from biogas upgrading or an external source; Rivera-Tinoco et al. [26] evaluated the methanol production cost for different types of used water electrolyzers.

Additional studies were about the methanol reactor. Generally, membrane reactor, adiabatic reactor in multi-bed configuration with inter-stage heat exchange (indirect cooling) and multi-tubular reactor (Lurgi type), where the reaction heat is removed by cooling with boiling water circulating externally to the reaction tubes, are considered [27].

Atsonios et al. [28] developed a reactor with membranes permeable to water and methanol, where  $CO_2$  is obtained by a power plant while  $H_2$  by water electrolysis in alkaline cells. Results showed that the use of membranes can increase the methanol yield, reducing then the recycling gas flow rate, the size of reactor and the investment cost. The methanol production cost was equal to 859.2 €/ton.

Gallucci et al. [29] studied experimentally a zeolite membrane reactor for methanol production. With a  $H_2/CO_2$  ratio in the feed equal to 3 and 7, the membrane reactor ensured higher methanol selectivity and yield compared to a traditional reactor.

Gallucci and Basile [30] modelled a membrane reactor for methanol production by  $CO_2$  and  $H_2$  on commercial catalyst using Graaf et al. [31] kinetic model; they compared the results with those obtained from a traditional reactor. It was found that  $CO_2$  conversion and methanol selectivity were higher in the membrane reactor compared to those in a traditional reactor, at the same experimental conditions.

Fornero et al. [32] studied a methanol reactor at different  $H_2/CO_2$  ratios ranging from 40/60 to 25/75: higher efficiencies were obtained at higher ratios, because a relatively higher amount of hydrogen was present.

Meyer et al. [33] modelled a methanol reactor (Lurgi type) with recycle of unconverted gases ( $CO$ ,  $H_2$ ,  $CO_2$ ) in MATLAB comparing the kinetic models of Graaf et al. [31] and Van den Bussche and Froment [24]. Results showed that both kinetic models have similar results when limitations by thermodynamic equilibrium are strong. However, differences in reaction rates were observed as far as the axial profiles of product concentrations along the reactor were concerned.

From the above literature analysis it is evident that a thorough comparison between different process schemes to produce methanol is lacking: there is ambiguity in the criteria to select among different processes. In addition, an equilibrium analysis of methanol reactor with pure  $CO_2$  and  $H_2$  in the feeding stream is not available. In this work we will supply these missing features.

In fact, the methanol production from pure  $CO_2$  and  $H_2$  was considered and a methodology was developed that helps comparison and decision-making for different process schemes. In particular, the process schemes of Perez-Fortes et al. [21] and Kiss et al. [18], both with the recycle of unconverted gases, were compared and analyzed in ChemCad®.

A thorough thermodynamic analysis was carried out for three different and novel reactor configurations using Aspen Plus®: once-through reactor, reactor with recycle of unconverted gases after condensation of methanol and water, reactor with membranes permeable to water. In the feed stream, only carbon dioxide and hydrogen were considered, differently from the majority of works analyzed in this chapter, which assume syngas in the feed stream. Although equilibrium conversion of  $CO_2$  may be different from experimental evidence and its simulation, both clearly affected by reaction rates, these calculations are important to “set the scene”, so to highlight intrinsic limitations that characterize the system. Their knowledge helps quantifying the effectiveness of different reactor configurations under real conditions, such as the recycle ratio of reactants, selective separation of reaction products, etc.

The reaction extent at thermodynamic equilibrium conditions and as a function of temperature was then compared with the conversion – temperature trajectory for an adiabatic reactor. The technical feasibility of methanol production from hydrogen and carbon dioxide was also demonstrated, showing kinetic and thermodynamic limits for different process schemes and comparing results obtained with different

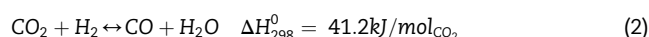
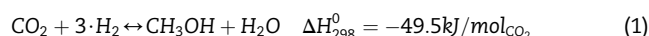
kinetic mechanisms and reaction rates proposed in the literature. In this way, a potential pathway of industrial interest for the utilization of  $CO_2$  emissions can be evaluated, after its capture from power plants. Finally, the simulation in MATLAB of a Lurgi type reactor was developed considering convective and axial dispersion flow.

Summarizing, the objectives of this work are: i) to proof the thermodynamic feasibility of methanol production with pure  $CO_2$  hydrogenation, ii) to develop and apply a methodology for the evaluation of different process schemes with recirculation of gas reactants, iii) to establish the optimal sizing of methanol reactor by means of a simulation model. The main scientific issue addressed in this work was indeed the interconnection existing among these three examined aspects (thermodynamics, plant configuration, reactor performance) of the system at hand.

## Materials and methods

### Kinetic model for methanol production

Different kinetic models were proposed to describe the methanol production from  $CO_2$  and  $H_2$  [33]. The kinetic models by Van den Bussche and Froment [24] and Graaf et al. [31] are most frequently used. The VdB-F (Van den Bussche and Froment) reaction mechanism assumes that  $CO_2$  hydrogenation produces methanol (stoichiometric Eq. (1)), as well as  $CO$ , the latter by means of the reverse Water Gas Shift (rWGS) that takes place simultaneously to methanol synthesis (stoichiometric Eq. (2)):



When the reactor feed contains  $CO_2$ ,  $H_2$  and  $CO$  (syngas), steam produced by reaction 1 may react with  $CO$  to obtain  $CO_2$ , made available for hydrogenation to methanol.

On the other hand, the Graaf reaction mechanism assumes that  $CO$  and  $CO_2$  hydrogenation are both involved in methanol synthesis; besides these two reactions, WGS also takes place.

As already mentioned in the Introduction, these two kinetic models may provide different product yield as a function of catalyst weight, although these differences become quite small when equilibrium conversion is approached (i.e. towards the exit of the reactor), so that both of them are usually considered satisfactory for hydrogenation to methanol utilizing syngas [33].

As stressed in the work by Van den Bussche and Froment [24], the mechanistically sound foundations of their model allow for accurate predictions of catalyst behavior outside the original experimental window, which in any case involved a quite large range of  $CO_2$  concentration in the reactor feed stream. Therefore, the VdB-F kinetic model was chosen for methanol reactor simulations under non-equilibrium conditions: it relates methanol and byproducts formation directly to carbon dioxide, the key reactant in the conditions of interest here.

The methanol reaction is exothermic with a reduction in the mole number, as shown by the stoichiometric expression

in eq. (1), while the reverse water gas shift (rWGS) is an endothermic reaction without change in the mole number, i.e. relatively high pressure levels are utilized to obtain higher conversions [34].

The respective reaction rate expressions are the following (see Eqs. (3) and (4)):

$$r_1 = \frac{k_1 \cdot p_{\text{CO}_2} \cdot p_{\text{H}_2} \cdot \left(1 - \frac{p_{\text{CH}_3\text{OH}} \cdot p_{\text{H}_2\text{O}}}{K_{\text{eq},1} \cdot p_{\text{H}_2}^3 \cdot p_{\text{CO}_2}}\right)}{\left(1 + k_c \cdot \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} + k_a \cdot \sqrt{p_{\text{H}_2}} + k_b \cdot p_{\text{H}_2\text{O}}\right)^3} \quad (3)$$

$$r_2 = \frac{k_2 p_{\text{CO}_2} \cdot \left(1 - \frac{p_{\text{CO}} \cdot p_{\text{H}_2\text{O}}}{K_{\text{eq},2} \cdot p_{\text{H}_2} \cdot p_{\text{CO}_2}}\right)}{\left(1 + k_c \cdot \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} + k_a \cdot \sqrt{p_{\text{H}_2}} + k_b \cdot p_{\text{H}_2\text{O}}\right)} \quad (4)$$

Partial pressures may be corrected by the gas fugacity coefficients, evaluated according to the Redlich-Kwong-Soave (RKS) equation of state [35]. In previous literature sources was found that, under temperature and pressure conditions of interest for this work, fugacity coefficients are practically equal to 1 [24,33].

The equilibrium constants (see Eq. 1 and 2) as functions of temperature are given by the following relations, respectively [24] (see Eqs. (5) and (6)):

$$\log_{10} K_{\text{eq},1} = \frac{3066}{T} - 10.592 \quad (5)$$

$$\log_{10} K_{\text{eq},2} = \frac{-2073}{T} + 2.029 \quad (6)$$

Generally, Cu/ZnO/Al<sub>2</sub>O<sub>3</sub>, a commercial catalyst, is used [16,36–39]: it is active at temperatures higher than 493 K, which means that the reaction conditions are limited by kinetic constraints.

### Modeling of methanol plants

Two process schemes for a methanol plant were considered and analyzed in ChemCad<sup>®</sup>: Perez-Fortes et al. [21] and Kiss et al. [18]. In both processes the recycle of unconverted gases (CO, CO<sub>2</sub>, H<sub>2</sub>) is present, after separation of methanol and water. Fig. 1a shows the scheme of Perez-Fortes et al. [21], while Fig. 1b shows the scheme of Kiss et al. [18].

In the process of Perez-Fortes et al. [21], the carbon dioxide feeding stream, captured from a pulverized coal power plant, is compressed through four stages of compressors with intermediate cooling from 1 bar to 78 bar, before to be mixed with compressed hydrogen at 78 bar. The methanol reactor was modelled as an adiabatic ideal plug flow reactor, using the kinetics of Van den Bussche and Froment [24]. The reactor operates at 76 bar and 483 K (inlet pressure and temperature) with a volume equal to 42 m<sup>3</sup> and with 44500 kg of Cu/ZnO/Al<sub>2</sub>O<sub>3</sub> catalyst arranged in pellets forming a packed bed. A heat exchanger removes the heat of reaction producing saturated steam, which expands into two turbines to generate electricity for the CO<sub>2</sub> compressors.

The remaining heat is used to heat cycled water that comes from the condenser of a power plant, assuming that the analyzed process is integrated into the power plant. Also, the stream at the exit of the reactor is used to pre-heat the feed

stream, to provide heat at the bottom of the distillation column and then to pre-heat the feeding stream to that column. After this thermal integration, methanol and water are separated and the unconverted gases are partially recirculated. The plant processes 80.5 ton/h of carbon dioxide and 11 ton/h of hydrogen, while 55.1 ton/h of methanol are produced in the distillation section. The carbon dioxide conversion in the reactor is equal to 21.97%, while carbon dioxide conversion in the whole plant is equal to 93.85%.

In the process of Kiss et al. [18] the fresh CO<sub>2</sub> feeding stream is mixed with the recycle gas stream and sent to the feed-effluent-heat exchanger without decompression and heating. This ensures a lower gas flow rate to the recycle compressor reducing the required power. The fresh, wet hydrogen feed from chlorine production by salt electrolysis is pressurized to 45 bar before entering the stripper and then is fed to the reactor. The methanol reactor is a multi-tubular reactor with the following characteristics: 12 m of length, 810 tubes of 0.06 m of diameter containing Cu/ZnO/Al<sub>2</sub>O<sub>3</sub> catalyst. It works in isothermal mode at 523 K and 50 bar. In Ref. [16] it was simulated using the kinetic model of Graaf et al. [31], while in this work the alternative kinetic mechanism and reaction rates proposed by Van den Bussche and Froment [24] (the VdB-F model) were adopted, as already mentioned. The heat of reaction could be used to produce high pressure steam. The stream at the outlet of the reactor is cooled by two heat exchangers and then flashed to separate gas (CO, CO<sub>2</sub>, H<sub>2</sub>) from the methanol-water mixture. The gas stream, after a tiny purge, is recirculated to the reactor, following compression and mixing with fresh reactants. The methanol-water mixture is sent to the stripper where wet hydrogen is fed in counter-current mode. This dries the hydrogen feed thus removing water from the reactor feed, and separates light gases (as H<sub>2</sub>, CO and CO<sub>2</sub>) that are completely recycled. The liquid bottom stream of the stripper is sent to a distillation column for the separation of methanol. 17209 kg/h of CO<sub>2</sub> and 2365 kg/h of H<sub>2</sub> are entering the plant, while 12507 kg/h of methanol are produced. Carbon dioxide conversion in the reactor is equal to 17.2%, while carbon dioxide conversion in the whole plant is equal to 100%.

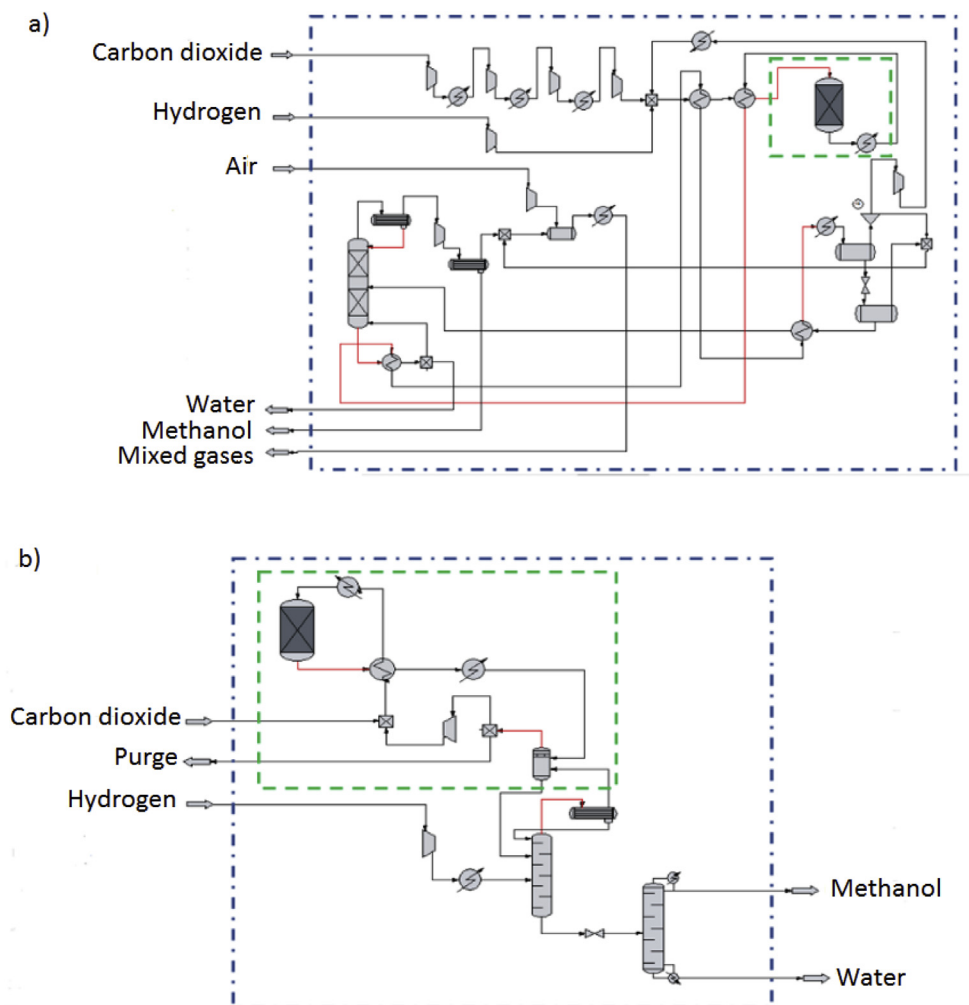
To simulate the kinetic aspects easily, the two plants were modelled in ChemCad<sup>®</sup> using RKS thermodynamic model for the plant studied by Perez-Fortes et al. [21] and Non Random Two Liquid (NRTL) thermodynamic model for the plant studied by Kiss et al. [18], following the choice made in those respective papers.

### Reactor configurations

The equilibrium analysis was developed considering three different reactor configurations: once-through reactor, as in Fig. 2a, reactor with recycle of unconverted gases after separation of methanol and water, as in Fig. 2b, and membrane reactor with separation of water during the reaction, as in Fig. 2c.

These were considered equilibrium reactors, characterized by the chemical reactions described with the stoichiometric expressions reported in Eqs (1) and (2) and respective equilibrium constants as in Eqs. (5) and (6). Their modeling was developed using Aspen Plus<sup>®</sup> software with RKS as





**Fig. 1 – Process scheme for methanol production according to: a) Perez-Fortes et al. [21], b) Kiss et al. [18] (green dashed line: boundaries of the reactor section; blue dashed line: boundaries of the plant section). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)**

thermodynamic model. The feeding stream is composed by  $\text{CO}_2$  and  $\text{H}_2$  in stoichiometric ratio.

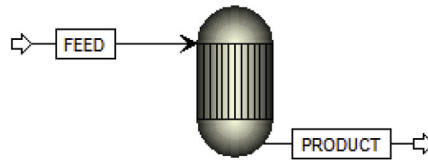
The membrane reactor was modelled using six stages of equilibrium reactors and water separators in order to simulate the simultaneous reaction and water separation process. In this case we were simulating a system operating under thermodynamic equilibrium conditions, so that the final result obtained is expected to be quite independent from the process path (arrangement of reaction step and simultaneous steam separation into a sequence of subsequent operations from among the software standard units). Water flowing through the membrane is considered to be a fraction,  $\alpha$ , of its mass in the reaction ambient. In particular,  $\alpha = 0$  for a reactor without steam separation, while  $\alpha = 1$  when membrane is considered to be able to extract from the reaction ambient all the water produced there. We assumed  $\alpha = 0.4$  in our calculations and checked that the driving force (difference between steam partial pressure in the retentate and permeate side, respectively) was always positive.

### Mathematical modeling of methanol reactor

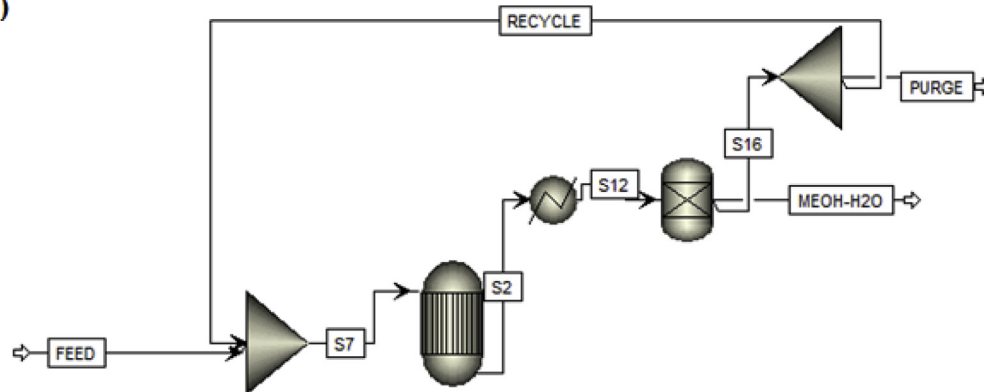
A mathematical model in steady state conditions was developed in MATLAB with the solver ode 45 for a methanol synthesis reactor with recycle of gases ( $R = 90\%$ ), according to the following assumptions [11]:

- negligible axial dispersion;
- maximum radial mixedness, (concentration, velocity and temperature profiles are assumed constant as functions of the reactor tube radius) due to the small ratio of reactor diameter to reactor length, leading to a one-dimensional model;
- pseudo-homogeneous and uniform catalytic particles: constant temperature and pressure profiles within the catalytic pellet due to the small size of the pellet and relatively low chemical reaction intrinsic rate at the moderate temperature level of the reactor: the Weisz-Prater parameter, defined as follows (see Eq. (7)) [40]:

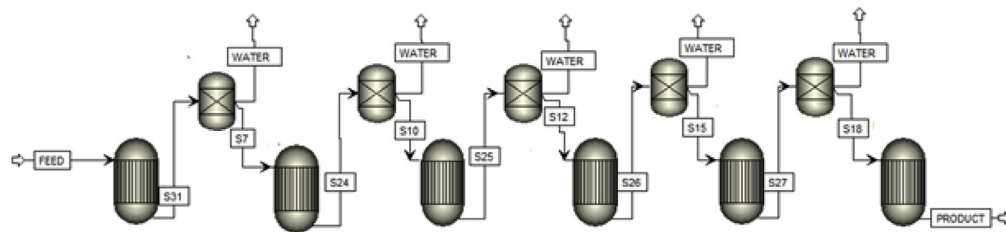
a)



b)



c)



**Fig. 2 – Methanol reactor configurations developed in Aspen Plus: a) once-through equilibrium reactor; b) equilibrium reactor with recycle of CO–CO<sub>2</sub>–H<sub>2</sub> and separation of CH<sub>3</sub>OH–H<sub>2</sub>O by condensation; c) membrane reactor with water separation.**

$$C_{wp} = \frac{r_{obs,i} \cdot \rho_c \cdot R^2}{D_e \cdot C_{s,i}} \quad (7)$$

is estimated to be  $\ll 1$  ( $C_{wp} \approx 0.03$ ); where  $r_{obs,i}$  is the measured value of the rate of reaction for component  $i$ ,  $C_{s,i}$  is the concentration at the pellet surface of component  $i$ ,  $D_e$  is the effective diffusivity,  $R$  is the pellet radius,  $\rho_c$  is the density of catalyst particle;

- negligible catalyst deactivation: because temperature is lower than levels where sintering phenomena may occur

(the life cycle of the commercial catalyst is in the order of two years and no sulfur species are considered in this case);

- negligible side reactions except for rWGS, because of high catalyst selectivity;
- pseudo-homogeneous model: gradients between gas and solid phase are neglected due to small size of particles;

The model is composed of the material balances, the energy balance and the Ergun equation to estimate pressure drop along the reactor. The recycle of gases (CO, CO<sub>2</sub>, H<sub>2</sub>), with a recycle ratio equal to 90%, after the separation of methanol

and water by condensation, is taken into account in the boundary conditions, which establish molar flows and concentrations entering the reactor by adding together the plant feeding stream (composed only by CO<sub>2</sub> and H<sub>2</sub>, with molar ratio equal to 1:3) and the recycle stream. The material balances for each component in steady state conditions assume that the variation of the convective molar flow of each chemical compound, at increasing mass of catalyst in the packed bed, matches the corresponding reaction rate (see Eqs. (8)–(12)):

$$\frac{dF_{CH_3OH}}{dw} = r_{CH_3OH} = r_1 \quad (8)$$

$$\frac{dF_{H_2O}}{dw} = r_{H_2O} = r_1 + r_2 \quad (9)$$

$$\frac{dF_{CO_2}}{dw} = r_{CO_2} = -r_1 - r_2 \quad (10)$$

$$\frac{dF_{CO}}{dw} = r_{CO} = r_2 \quad (11)$$

$$\frac{dF_{H_2}}{dw} = r_{H_2} = -3 \cdot r_1 - r_2 \quad (12)$$

where  $F_i$  is the molar flow rate of  $i$  component (CH<sub>3</sub>OH, H<sub>2</sub>O, CO<sub>2</sub>, CO, H<sub>2</sub>) in mol/s,  $r_1$  and  $r_2$  are the reaction rates for methanol and carbon monoxide formation, respectively, referred to reactions (1) and (2) and expressed by Eqs. (3) and (4), in mol/kg s,  $w$  the mass of catalyst in kg. The composition profile for each component is then obtained. A diffusion-like term was added in the above equations in order to highlight the influence of axial dispersion. The energy balance and Ergun equations were used to calculate the temperature and pressure profiles, respectively from (see Eqs. (13)–(16)):

$$\frac{dT}{dw} = \frac{U \cdot a \cdot (T_j - T)}{F \cdot C_{pm}} - \frac{\Delta H_1^{rxn} \cdot r_1}{F \cdot C_{pm}} - \frac{\Delta H_2^{rxn} \cdot r_2}{F \cdot C_{pm}} \quad (13)$$

$$\frac{dP}{dw} = -\frac{G}{\rho \cdot g_c \cdot d_p} \cdot \left( \frac{1 - \phi}{\phi^3} \right) \cdot \left[ \frac{150 \cdot (1 - \phi) \cdot \mu}{d_p} + 1.75 \cdot G \right] \cdot \frac{z}{w} \quad (14)$$

$$\frac{z}{w} = \frac{1}{S \cdot (1 - \phi) \cdot \rho_p} \quad (15)$$

$$a = \frac{4}{D_t \cdot (1 - \phi) \cdot \rho_p} \quad (16)$$

where  $F$  is the total molar flow rate,  $w$  is the mass of catalyst in kg,  $T$  is the temperature in K,  $U$  is the global coefficient of heat exchange in J/m<sup>2</sup>s K,  $a$  is the specific surface area of heat exchange in m<sup>2</sup>/kgcat,  $T_j$  is the temperature of the heat exchange fluid in K,  $C_{pm}$  is the average specific heat in J/molK,  $\Delta H^{rxn}$  is the heat of reaction in J/mol,  $r$  is the reaction rate for reactions 1 and 2, respectively, in mol/kg s (see Eqs. (3) and (4)),  $P$  is the pressure in Pa,  $G$  is the superficial mass velocity in kg/s m<sup>2</sup>,  $\rho$  is the gas density in kg/m<sup>3</sup>,  $g_c$  is a conversion factor in kg/m/s<sup>2</sup>N,  $d_p$  is the average equivalent diameter of catalyst pellets in the bed in m,  $\phi$  is the bed porosity,  $\mu$  is the fluid viscosity in Pa s,  $z$  is reactor length in m,  $D_t$  is the tube diameter,  $S$  the tube cross section surface area,  $\rho_p$  is the density of catalyst pellets. The

axial dispersion coefficient was taken into account when considering the dispersion term in the balance equations.

The energy balance (Eq. (13)) relates the temperature variation along the bed of catalyst to the heat transferred from/to the heat exchange medium and to that produced/absorbed by reactions 1 and 2. On the other hand, Eq. (14) (Ergun equation) estimates the pressure drop through a linear combination of mechanical energy dissipation due to creeping and turbulent flow. As mentioned above, in the reactor model developed here the Van den Bussche and Froment [24] kinetic model was chosen to provide the constitutive equations needed to compute the chemical reaction rates at different conditions. The reactor is a multi-tubular reactor with heat exchange at the tube wall, Lurgi type, and containing Cu/ZnO/Al<sub>2</sub>O<sub>3</sub> as catalyst.

## Results and discussions

### Results of equilibrium analysis

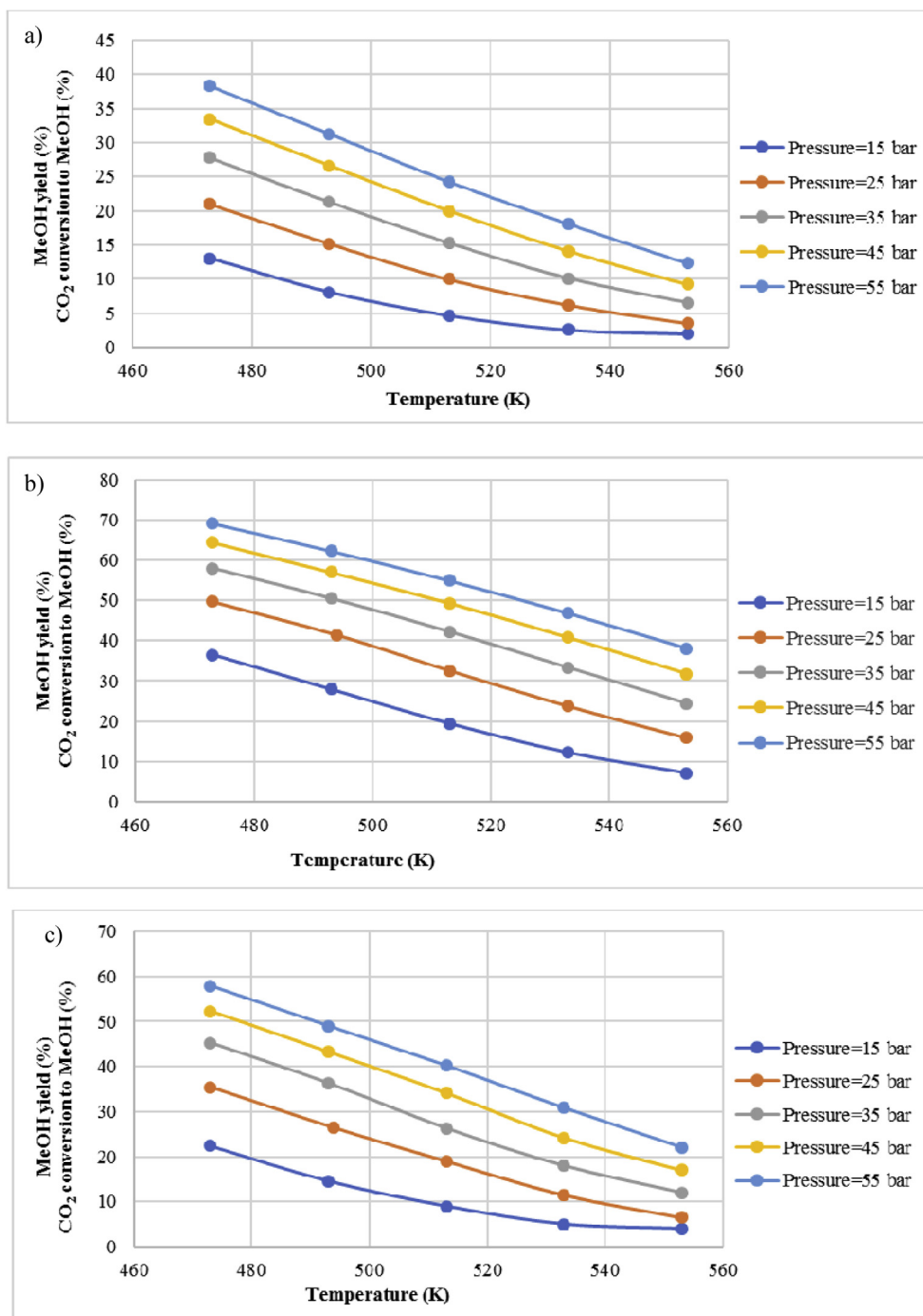
#### Equilibrium methanol reactor

A thermodynamic equilibrium analysis of a methanol reactor in different plant configurations, as i) once-through reactor, ii) reactor with recycle of unconverted gases after separation of methanol and water ( $R = 80\%$ ) and iii) membrane reactor, was developed in this section with the aim to evaluate the CO<sub>2</sub> conversion to methanol, i.e. methanol yield, CO<sub>2</sub> conversion to CO, carbon dioxide global conversion, methanol selectivity. For the reactor with recycle, the recycle stream is included in the system boundaries, as shown in Fig. 2. In this analysis, the range of temperature was fixed to be 473–553 K, where the industrial catalyst based on Cu/Al<sub>2</sub>O<sub>3</sub>/Zn would assure satisfactory rates of reaction and thermodynamic equilibrium of reaction 1 is also quite favorable. The commonly suggested range of operating pressure is 15–55 bar. The external, fresh feed stream to the plant contains 1:3 mol/h of CO<sub>2</sub> and H<sub>2</sub>, according to stoichiometric conditions for reaction 1.

Fig. 3 shows that temperature has a negative effect, while pressure has a positive effect on carbon dioxide conversion to methanol according to the Le Chatelier's principle (an exothermic reaction with reduction of mole number).

More specifically, Fig. 3a shows carbon dioxide conversion to methanol and methanol yield varying temperature and pressure, for once-through reactor. The highest carbon dioxide conversion is reached at the lowest examined temperature (473 K) and highest examined pressure (55 bar). The results of temperature-dependent analysis clearly show the limitations of direct CO<sub>2</sub> hydrogenation route, since the maximum attainable CO<sub>2</sub> equilibrium conversion is 38% at the conditions considered here.

For the reactor with a recycle of gases ( $R = 80\%$ ) after separation of methanol and water, Fig. 5b shows that, at 55 bar and 473 K, carbon dioxide conversion to methanol and methanol yields are equal to 69%, with reference to the plant feed stream. Definitely, the system with a recycled stream of H<sub>2</sub>–CO<sub>2</sub>–CO, after separation of methanol and water, ensures a higher methanol yield compared to the process arrangement with a mixture of CO<sub>2</sub>–H<sub>2</sub> in the reactor feed stream, examined before.



**Fig. 3** – Equilibrium CO<sub>2</sub> conversion to methanol and methanol yield as functions of temperature and pressure, with stoichiometric CO<sub>2</sub>–H<sub>2</sub> ratio (1:3) in the process feeding stream: a) once-through reactor; b) reactor with recycle of gases (R = 80%) after separation of methanol and water; c) reactor with membrane permeable to steam.

Fig. 3c shows the equilibrium analysis for a membrane reactor designed to separate water simultaneously to the reaction process. The system ensures a CO<sub>2</sub> conversion higher than the one in a once-through reactor, but substantially lower when compared to CO<sub>2</sub> conversion predicted in a reactor with recycle of gases after separation of condensable compounds (Fig. 3b). In fact, at the same operating conditions, 55 bar and 493 K, the equilibrium CO<sub>2</sub> conversion to methanol is equal to 58%. When compared to results obtained with

once-through reactor (Fig. 3a), the higher conversion is due to the removal of water that shifts the reactions towards products. It is worth noticing here that H<sub>2</sub>O is also produced by rWGS reaction, so selectivity towards methanol is not helped.

The positive effect on the conversion process brought about by water removal is also reported by Zachopoulos and Heracleous [41].

The results reported in Fig. 3 show that by combining the separation of reaction products, as water and methanol, and



by recycling unconverted gases, CO<sub>2</sub> conversion and methanol yield can be improved compared to other cases. A similar behavior was also reported in studies by Makertiharta et al. [42], Struis and Stucki, [43]. The recycle of unconverted gases allows to overcome the equilibrium restrictions on product yield referred to the plant feed stream composition, and to increase CO and H<sub>2</sub> in the reactor feed stream besides CO<sub>2</sub>.

It is clear that a continuous removal of products can improve the methanol yield and may allow reducing the synthesis pressure [43], as a consequence of the law of mass action.

The presence of CO in the feeding syngas is highly advantageous with respect to the methanol synthesis from plain CO<sub>2</sub>. According to van den Bussche and Froment [24] reaction mechanism, increasing CO concentration makes the equilibrium extent of the second reaction (rWGS) to decrease progressively up to the point when it proceeds in the opposite direction, when CO concentration is higher than that at equilibrium, and in these conditions WGS contributes to increase CO<sub>2</sub> availability and therefore methanol yield. Water inhibits methanol synthesis and a further positive effect of CO

arises from its ability to “scavenge” excess water via water gas shift. Then, recycle of CO in the feed to the reactor allows to obtain a higher methanol yield and carbon conversion at thermodynamic equilibrium. As a matter of facts, Fig. 4 shows the methanol yield and carbon conversion to methanol versus the CO/CO<sub>2</sub> ratio in the feed and the pressure level, at different temperature (the feeding concentration of hydrogen is adjusted in each case according to the stoichiometric conditions). It is evident that a higher amount of CO in the feed allows to obtain a higher methanol yield and carbon conversion at thermodynamic equilibrium. Temperature has a negative effect while pressure has a positive effect also on the cases analyzed here. A value of methanol yield and carbon conversion near to 100% can be obtained only at the lowest temperature (473 K) investigated here, while at the highest temperature (553 K) methanol yield and carbon conversion are equal to 39% at most.

A maximum in methanol production as a function of CO<sub>2</sub>/CO ratio occurs at CO<sub>2</sub> concentration equal to 2–5 mol% [24,44]. As reported also by Skrzypek et al. [45], if there is no CO in synthesis gas, the mole fraction of methanol in the

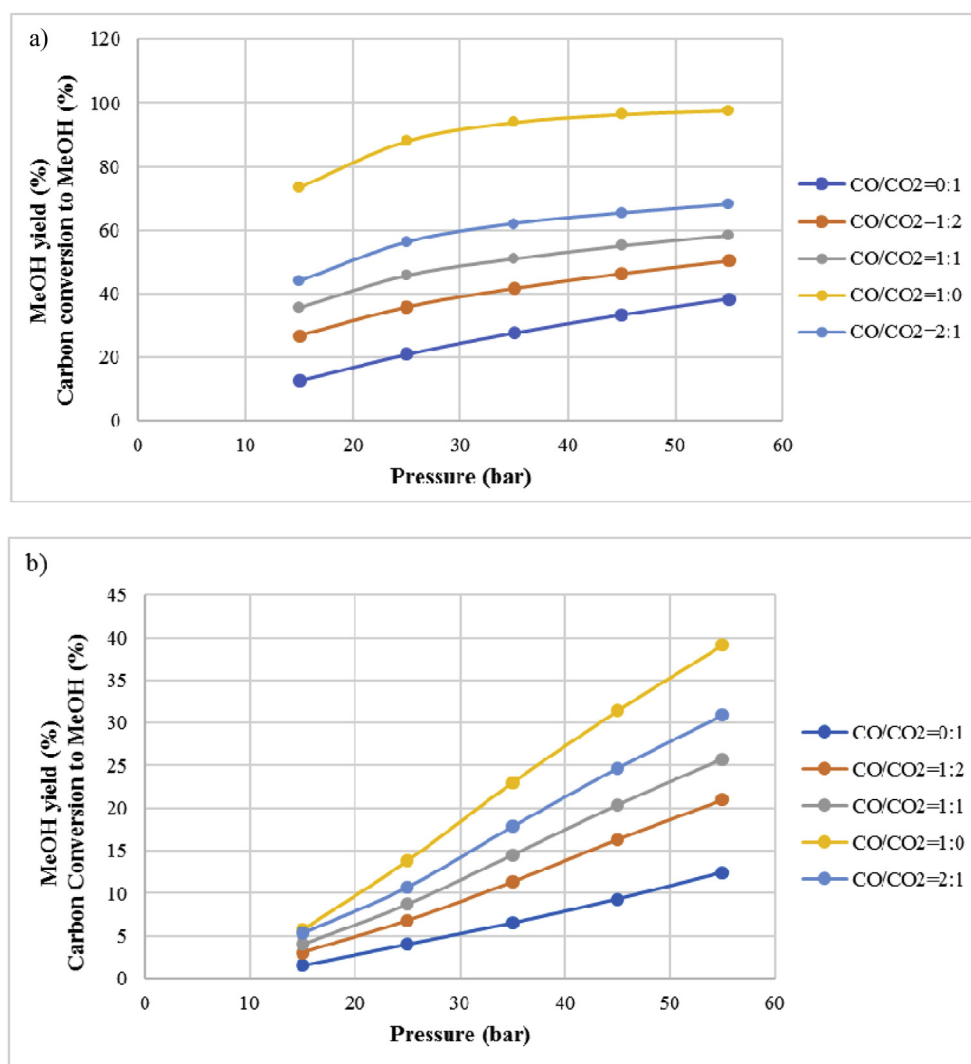


Fig. 4 – Equilibrium methanol yield and carbon conversion to methanol for once-through reactor, as functions of pressure, at different CO/CO<sub>2</sub> ratios, with H<sub>2</sub> content in the feed always adjusted to stoichiometry; a) T = 473 K; b) T = 553 K.

condensable products does not exceed 0.5, in agreement with the stoichiometry of reactions 1 and 2.

The positive effect of  $H_2$  content on process efficiency was assessed by additional thermodynamic calculations, not shown here. At 493 K and 55 bar, in the case of stoichiometric ratio 1:3, carbon dioxide conversion is equal to 38% (Fig. 3a), while at the same temperature and pressure carbon dioxide equilibrium conversion reaches 54% with a  $CO_2:H_2$  ratio equal to 1:5. The positive effect of  $H_2$  on  $CO_2$  conversion is also reported by Bansode and Urakawa [46]. However, the  $H_2$  conversion decreases with the increase of just  $H_2$  partial pressure, as expected when considering the excess of  $H_2$  relative to  $CO_2$ . Khalilpourmeymandi et al. [47] find that the effect of the relative amount of hydrogen on methanol production is higher than that of a relative amount of carbon monoxide.

The global  $CO_2$  conversion as a result of both, reaction 1 and reaction 2, was also analyzed at thermodynamic equilibrium, as shown in Fig. 5. The global  $CO_2$  conversion shows a minimum in the examined range of conditions under most of the pressure levels examined in this work. Comparing Figs. 5a and 3a, as well as for remaining reactor configurations, shows that, at lower temperature, reaction 1 mainly occurs, so the system reacts selectively to produce methanol. In fact, at 473 K and 55 bar, carbon dioxide conversion in Fig. 3a is equal to 38% while global carbon conversion in Fig. 5a is only 40%. However, at higher temperature, carbon dioxide reacts more extensively according to the reverse water gas shift (reaction 2) and produces methanol in minor quantity. In fact, at 553 K and 55 bar, carbon dioxide conversion in Fig. 5a is 13% while global carbon dioxide conversion in Fig. 9a is 25%. From Fig. 5 it is also evident that the global  $CO_2$  consumption in the reactor with recycle of gases reaches the highest values. In fact, at 493 K and 55 bar, once-through reactor, reactor with recycle of unconverted gases and membrane reactor have respectively a global  $CO_2$  conversion equal to 40%, 79%, 60%. Conversion of  $CO_2$  with respect to the plant feed stream is largely determined by the value chosen for the recycle ratio ( $R = 80\%$  in this case), when the reactor with recycle of unconverted gases is considered.

Summarizing the thermodynamic predictions obtained at different temperatures and pressures and with the three different reactor configurations examined here, the selectivity to methanol (reaction 1) decreases with increasing temperature, due to the favored endothermic reverse water gas shift reaction to CO (reaction 2). Pressure has a positive effect. At a temperature equal to 493 K and a pressure equal to 55 bar, methanol selectivity for once-through reactor, reactor with recycle and membrane reactor is equal respectively to 96%, 88% and 96%. Then the recycle reactor ensures higher  $CO_2$  conversion to methanol, although slightly lower methanol selectivity.

#### Adiabatic methanol reactor

For the examination of the performance of an adiabatic methanol reactor, the gases in the recycle stream were kept at the reactor inlet temperature, equal to 473 K, and pressure, equal to 55 bar. The external, fresh feed is composed by carbon dioxide and hydrogen in stoichiometric ratio. It was assumed that the reactor exit conditions approach those of equilibrium at 95%. The kinetic model of Van den Bussche and

Froment [24] for methanol and rWGS reactions was considered to describe the behavior of the adiabatic reactor. The extent of reaction 1 as a function of temperature and the corresponding equilibrium curve were obtained, as shown in Fig. 6.

Fig. 6a shows the temperature – conversion path of the adiabatic methanol reactor with 60% recycle ratio of gases after condensation of methanol and water. Temperature at the outlet of the real reactor is equal to 544 K in this case, while the methanol yield is equal to 27% with reference to  $CO_2$  in the fresh feed to the plant. Additional results are reported in Table 1.

Fig. 6b shows the behavior of the adiabatic reactor with 80% recycle ratio of gases after condensation of products (methanol and water). The temperature at the outlet of the adiabatic reactor is slightly different, equal to 547 K, while the extent of reaction 1, or methanol yield, is substantially higher, equal to 37%, as shown in Table 1.

Fig. 6c shows corresponding results obtained with 90% recycle of gases after separation of methanol and water by condensation. As shown in Table 1, the outlet temperature is 550 K, while the effective methanol yield is 45%.

These results show that an adiabatic methanol reactor with  $CO_2$  and  $H_2$  in the plant external feed allows for a higher methanol yield when increasing the recycle ratio, with a corresponding slight increment of the reactor outlet temperature, because of the increased gas flow rate through the reactor and associated heat capacity. It is worth noticing here that, with a process feed stream made of  $CO_2$  and  $H_2$ , the heat of reaction is lower compared to that with syngas in the feed: it is about half of the heat of reaction obtained by using syngas in the feed.

#### Results of methanol plant modeling

Modeling and analysis of methanol plants reported in Perez-Fortes et al. [21] and Kiss et al. [18] was carried out in ChemCad<sup>®</sup>. Table 2 shows that a fairly good agreement between literature and simulation data is present, so the models developed in this work were both validated. The plant of Perez-Fortes et al. [21] is operating at a much larger carbon dioxide feed as compared to the plant of Kiss et al. [16], therefore with a higher methanol throughput capacity: 55100 kg/h of methanol are produced with the former plant against 12507 kg/h with the latter one. The plant by Kiss et al. [18] is also characterized by higher carbon dioxide conversion and stoichiometric number in the reactor. In fact, in Perez-Fortes et al. [21] carbon dioxide conversion in the reactor is equal to 22% while in Kiss et al. [16] this value is equal to 27%. The stoichiometric number in the reactor in Perez-Fortes et al. [21] and Kiss et al. [18] is equal to 2.3 and 3.7, respectively: the higher value suggests a larger relative amount of hydrogen that increases the carbon dioxide conversion. However, the plant yield is higher in the scheme of Kiss et al. [18] due to a lower purge of gas reactants. For this reason, carbon dioxide conversion of the process is equal to 93.9% in Perez-Fortes et al. [21] and almost 100% in Kiss et al. [18]. In addition, the specific utilization of  $CO_2$  (as kg/kgMeOH) predicted by our calculations is very close for both plant schemes and matches perfectly that reported by the respective literature sources: it

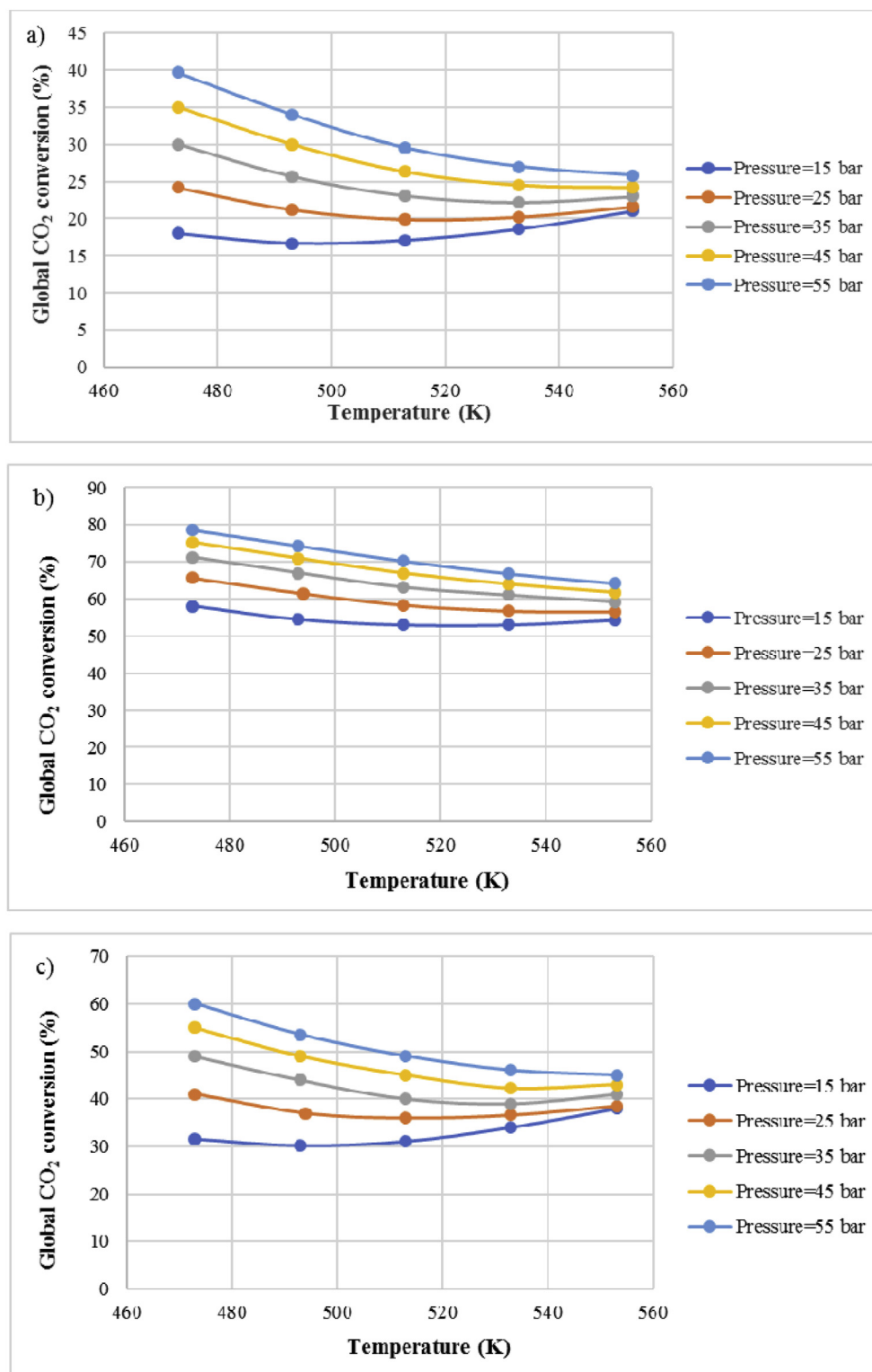


Fig. 5 – Overall CO<sub>2</sub> conversion at equilibrium conditions, as a function of temperature and pressure, with CO<sub>2</sub>–H<sub>2</sub> (1:3) ratio in the process feeding stream: a) once-through reactor; b) reactor with recycle of gases (R = 80%) after separation of methanol and water by condensation; c) reactor with membrane permeable to steam.

is equal to 1.38 kg/kgMeOH in Kiss et al. [18], while in Perez-Fortes et al. [21] is equal to 1.37 kg/kgMeOH.

Finally, it is worth recalling here that the reaction mechanism and reaction rates suggested by Van den Bussche and

Froment [24] were utilized in this work to obtain the results reported in Table 2, differently from the approach by Kiss et al. [18], who adopted the kinetic model by Graaf et al. [31]. A very close methanol throughput was predicted in our simulations,

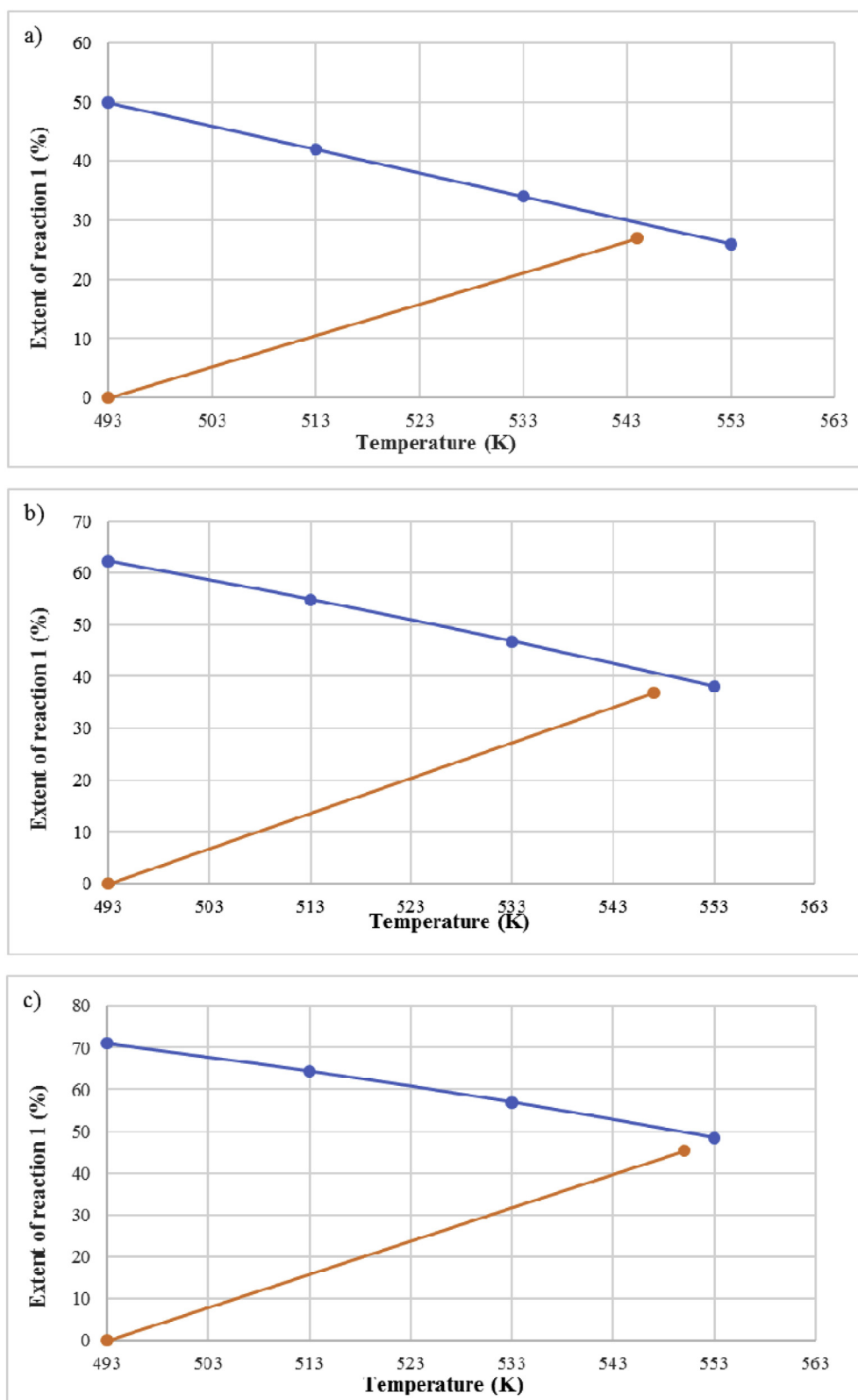


Fig. 6 – Extent of reaction 1-vs-temperature trajectory for the adiabatic methanol reactor with recycle of gases after separation of methanol and water by condensation (orange line), and corresponding thermodynamic equilibrium limit as a function of temperature (blue line): a)  $R = 60\%$ ; b)  $R = 80\%$ ; c)  $R = 90\%$ . (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

**Table 1 – Calculated performance indicators for the adiabatic methanol reactor and for the whole process, with different recycle ratio of gases after separation by condensation of methanol and water.**

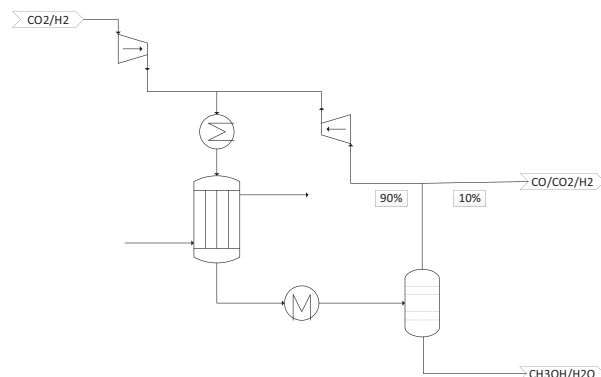
| Recycle ratio                                            | 60%   | 80%   | 90%   |
|----------------------------------------------------------|-------|-------|-------|
| CO <sub>2</sub> flow rate, at the reactor inlet (mol/h)  | 1.705 | 2.28  | 2.77  |
| CO <sub>2</sub> flow rate, at the reactor outlet (mol/h) | 1.306 | 1.78  | 2.19  |
| CO <sub>2</sub> flow rate, at plant inlet (mol/h)        | 1     | 1     | 1     |
| CO <sub>2</sub> flow rate, at plant outlet (mol/h)       | 0.522 | 0.365 | 0.219 |
| MeOH production (mol/h)                                  | 0.284 | 0.388 | 0.478 |
| Temperature, reactor inlet (K)                           | 493   | 493   | 493   |
| Temperature, reactor outlet (K)                          | 544   | 547   | 550   |
| CO <sub>2</sub> conversion, in the whole process (%)     | 47.8  | 63.5  | 78.1  |
| CO <sub>2</sub> conversion, in the reactor (%)           | 23.4  | 21.9  | 20.93 |
| MeOH yield, in the whole process (%)                     | 26.9  | 36.8  | 45.4  |
| MeOH yield, in the reactor (%)                           | 16.7  | 17    | 17.25 |
| MeOH selectivity, in the whole process (%)               | 59.4  | 61.1  | 61.2  |
| MeOH selectivity, in the reactor (%)                     | 71.2  | 77.6  | 82.4  |

with the same plant configuration, reactant feed streams and reactor size, as shown on Table 2. According to the VdB-F kinetic model, the recycle stream is somewhat reduced: this allows for a greater contact time with a consequent higher conversion of CO<sub>2</sub> in the reaction section of the whole plant (Table 2).

#### Results of simulation of methanol reactor

Simulation of the methanol reactor, Lurgi type [11], was developed in MATLAB software and was validated considering the work of Yusup et al. [48]. In the reactor outlet, water and methanol are separated by condensation while the unconverted gases (CO, CO<sub>2</sub>, H<sub>2</sub>) are recycled with a recycle ratio equal to 90% in order to improve methanol yield according to previous results, as is schematically shown in Fig. 7.

The operating conditions for the simulations are listed in Table 3. The external feed is composed of pure CO<sub>2</sub> and H<sub>2</sub> in stoichiometric ratio (the total feed flow rate is equal to 100 mol/s), while the feed to the reactor, following mixing with the recycle stream, has the relative composition H<sub>2</sub>/CO<sub>2</sub>/CO corresponding to the reactor performance recorded under each set of steady state operating conditions. The inlet temperature is 498 K while the inlet pressure is 55 bar. A Cu/Al<sub>2</sub>O<sub>3</sub>/Zn catalyst is considered, with the following characteristics: particle density 1175 kg/m<sup>3</sup>, pellet diameter 0.006 m, overall bed weight 2235 kg; the bed void fraction is fixed to 0.4. In the reactor, 150 tubes are assembled in parallel, with ID 0.058 m and length of 8 m, immersed in a shell where an isothermal

**Fig. 7 – Process scheme adopted in simulations for methanol reactor with 90% recycle of unconverted gases after separation of methanol and water.****Table 3 – Parameter values for simulation of methanol reactor in MATLAB software.**

|                                                        |           |     |     |
|--------------------------------------------------------|-----------|-----|-----|
| Inlet temperature (K)                                  | 498       |     |     |
| Inlet pressure (bar)                                   | 55        |     |     |
| Tube length (m)                                        | 8         |     |     |
| Tube ID diameter (m)                                   | 0,058     |     |     |
| Tube number                                            | 150       |     |     |
| Catalyst diameter (m)                                  | 0.006     |     |     |
| Bed void fraction                                      | 0.4       |     |     |
| Catalyst density (kg/m <sup>3</sup> )                  | 1175      |     |     |
| Catalyst weight (kg)                                   | 2235      |     |     |
| Isothermal heat exchange fluid temperature (K)         | 523       | 513 | 490 |
| Global heat transfer coefficient (J/m <sup>2</sup> sK) | 1000      | 600 | 300 |
| Process inlet flow rate (mol/s)                        | 100       |     |     |
| H <sub>2</sub> inlet flow rate (mol/s)                 | 75        |     |     |
| CO <sub>2</sub> inlet flow rate (mol/s)                | 25        |     |     |
| Recycle ratio (%)                                      | 90        |     |     |
| Reactor WHSV (1/h)                                     | 3.23–3.58 |     |     |

heat exchange fluid flows at different temperature levels (see Table 3).

According to these conditions a one dimensional model was developed in MATLAB as discussed in Section Mathematical modelling of methanol reactor, made of molar balances for each component, the energy balance and the pressure drop equation, with and without an axial dispersion term in the molar balance equations. The axial dispersion coefficient is fixed to 0.2 m<sup>2</sup>/s. Results show that, with the values considered for the system parameters, a slight

**Table 2 – Comparison of simulations in ChemCad® and literature data, for the process schemes proposed by Kiss et al. [18] and Perez Fortes et al. [21], respectively.**

|                                                                             | Perez-Fortes et al. (2016) | Kiss et al. (2016) | Simulation of Perez-Fortes et al. (2016) | Simulation of Kiss et al. (2016) |
|-----------------------------------------------------------------------------|----------------------------|--------------------|------------------------------------------|----------------------------------|
| Methanol production (kg/h)                                                  | 55100                      | 12507              | 57527                                    | 12492                            |
| Water production (kg/h)                                                     | 31700                      | 7346               | 31664                                    | 6065                             |
| CO <sub>2</sub> conversion (%), reactor/plant                               | 22.0/93.9                  | 17.2/100.0         | 22.0/96.5                                | 27/100.0                         |
| Gross CO <sub>2</sub> used (kg/kgCH <sub>3</sub> OH)                        | 1.37                       | 1.38               | 1.40                                     | 1.37                             |
| [H <sub>2</sub> ]-[CO <sub>2</sub> ]/[CO]+[CO <sub>2</sub> ], reactor/plant | 2.30/2.0                   | 1.64/2.00          | 2.31/2.00                                | 3.7/2.00                         |
| Heat flow due to reaction (MW)                                              | –23.3                      | –4.5               | –23.7                                    | –5.1                             |



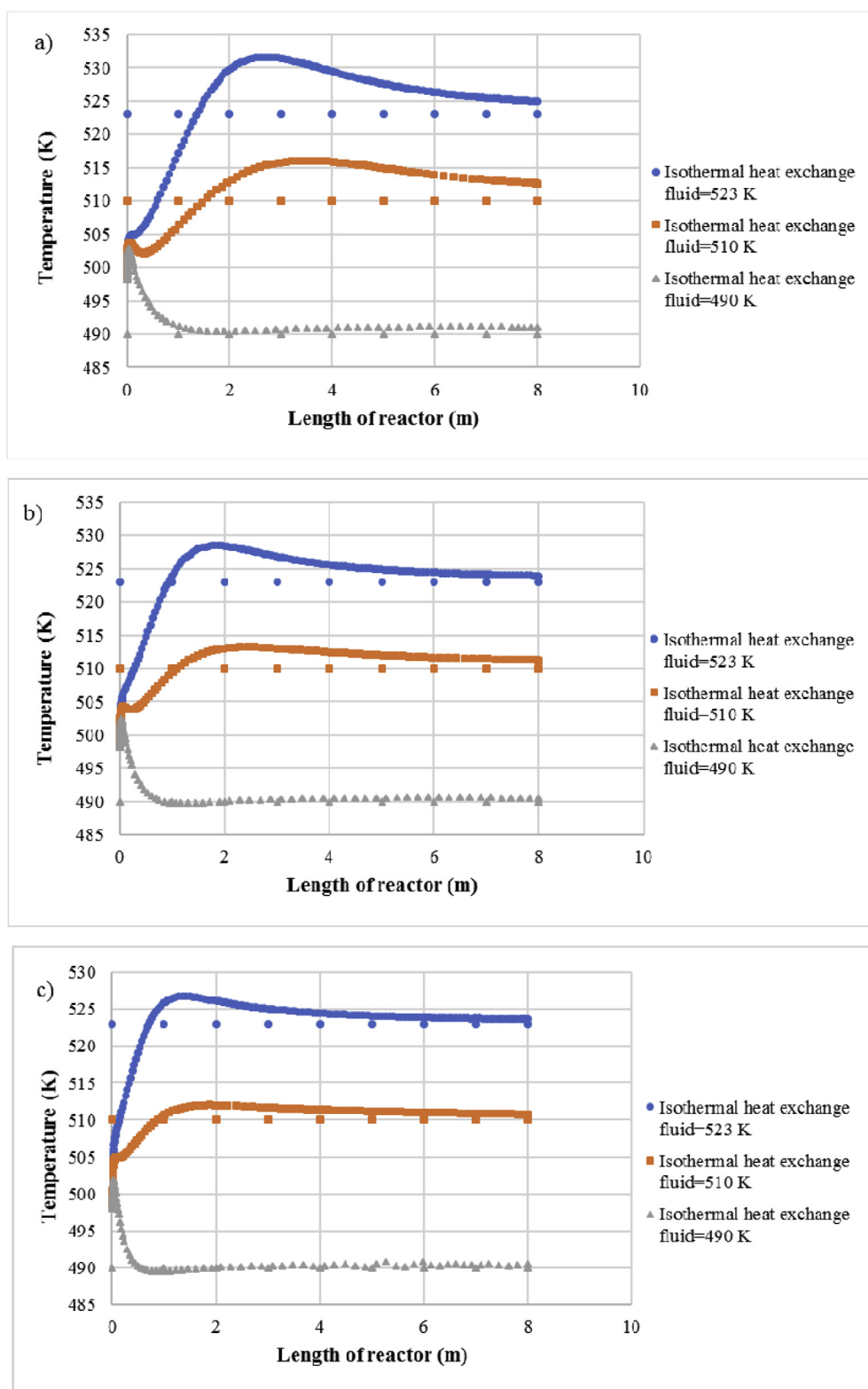


Fig. 8 – Calculated temperature profiles inside methanol reactor for different values of heat exchange fluid temperature and global heat exchange coefficient: a) 300 J/sm<sup>2</sup>K; b) 600 J/sm<sup>2</sup>K; c) 1000 J/sm<sup>2</sup>K.

difference is present between the two models, in agreement with the assumption made in different literature works suggesting to neglect axial dispersion in the methanol reactor [11,33,49–51]. A low pressure drop is obtained: the pressure at the outlet of the reactor is predicted to be equal to 54 bar.

The one-dimensional model is normally satisfactory for homogeneous reactors [52]: in packed beds, catalyst pellets hinder radial mixing, however it is worthwhile to consider such a simplified approach to obtain a reasonable prediction for the average temperature across the bed without the knowledge of effective thermal conductivity and diffusivity for heat and mass transfer. Therefore, this model is useful to investigate the main effects of different variables; it requires only a heat transfer coefficient at the wall of the reactor tubes. In the literature, different values for this coefficient are reported: 1000 J/sm<sup>2</sup>K by Bohn [53], 500 J/sm<sup>2</sup>K by Groppi et al. [54] and 250 J/sm<sup>2</sup>K by Luyben [55]. The presence of catalyst pellets increases the heat transfer coefficient at the wall of a packed bed considerably above that in an empty tube, at similar gas flow rate [56]. This large increase is due

to mixing and turbulence induced by particles and is quite difficult to predict it accurately as a function of process conditions, although some correlations have been proposed. For this reason, a sensitivity analysis is performed varying the value of global heat exchange coefficient over the whole range considered in the literature, and also varying the temperature of the isothermal heat exchange fluid according to Table 3.

Fig. 8 shows temperature axial profiles inside methanol reactor obtained with different values of the global heat exchange coefficient (300 J/sm<sup>2</sup>K in Fig. 8a and 600 J/sm<sup>2</sup>K in Fig. 8b and 1000 J/sm<sup>2</sup>K in Fig. 8c). In each figure, a value equal to 490 K, 510 K and 523 K is assumed for the isothermal heat exchange fluid, respectively. Results show that a value equal to 523 K for the heat exchange fluid ensures to have the best results in terms of reaction progress and selectivity, as is shown below. With reference to the thermodynamic equilibrium analysis discussed above and within the limits considered here, the dependence of chemical reaction rates on temperature, as proposed by Van den Bussche and Froment

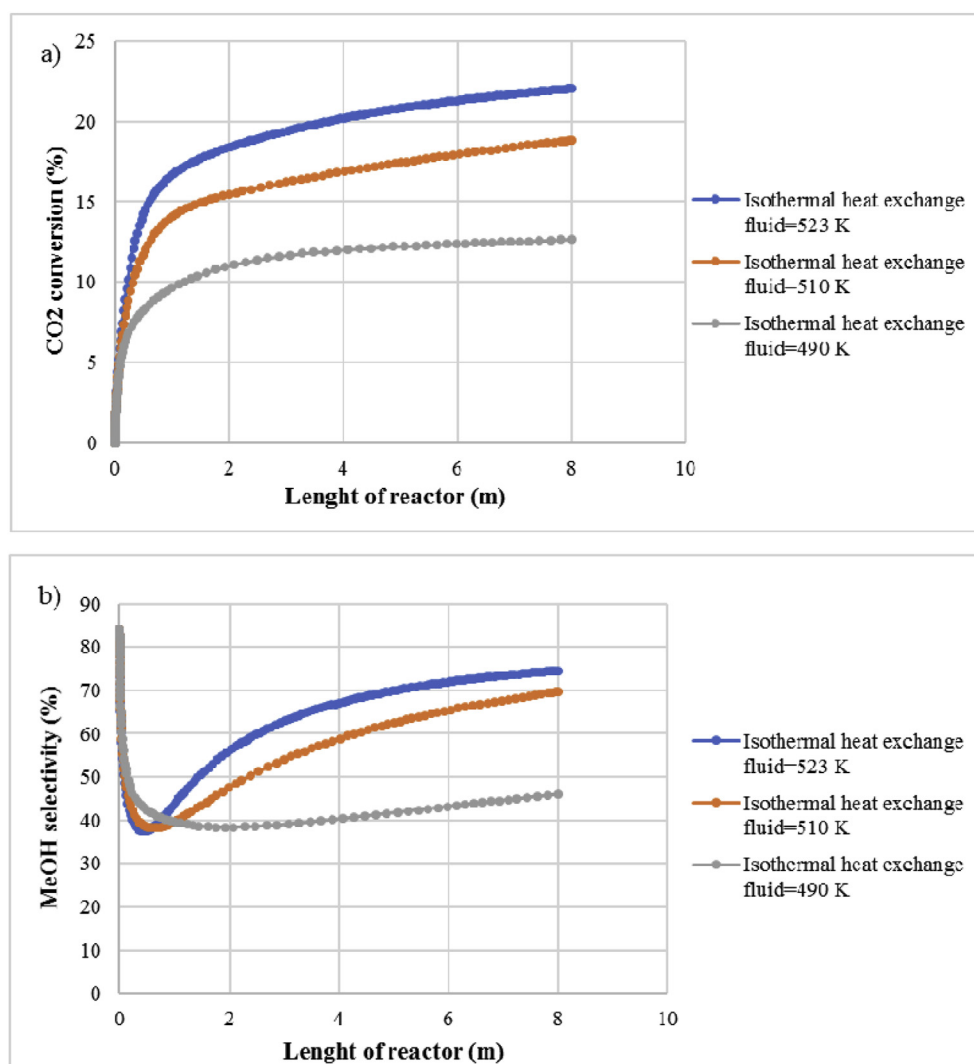


Fig. 9 – Additional numerical results obtained with a global heat exchange coefficient equal to 1000 J/sm<sup>2</sup>K and at different values of heat exchange fluid temperature: a) Overall carbon dioxide conversion in the reactor; b) Methanol selectivity.

**Table 4 – Calculated performance indicators for the nearly isothermal methanol reactor and the whole process with a value of global heat exchange coefficient equal to 1000 J/sm<sup>2</sup>K and at different temperature of isothermal heat exchange fluid.**

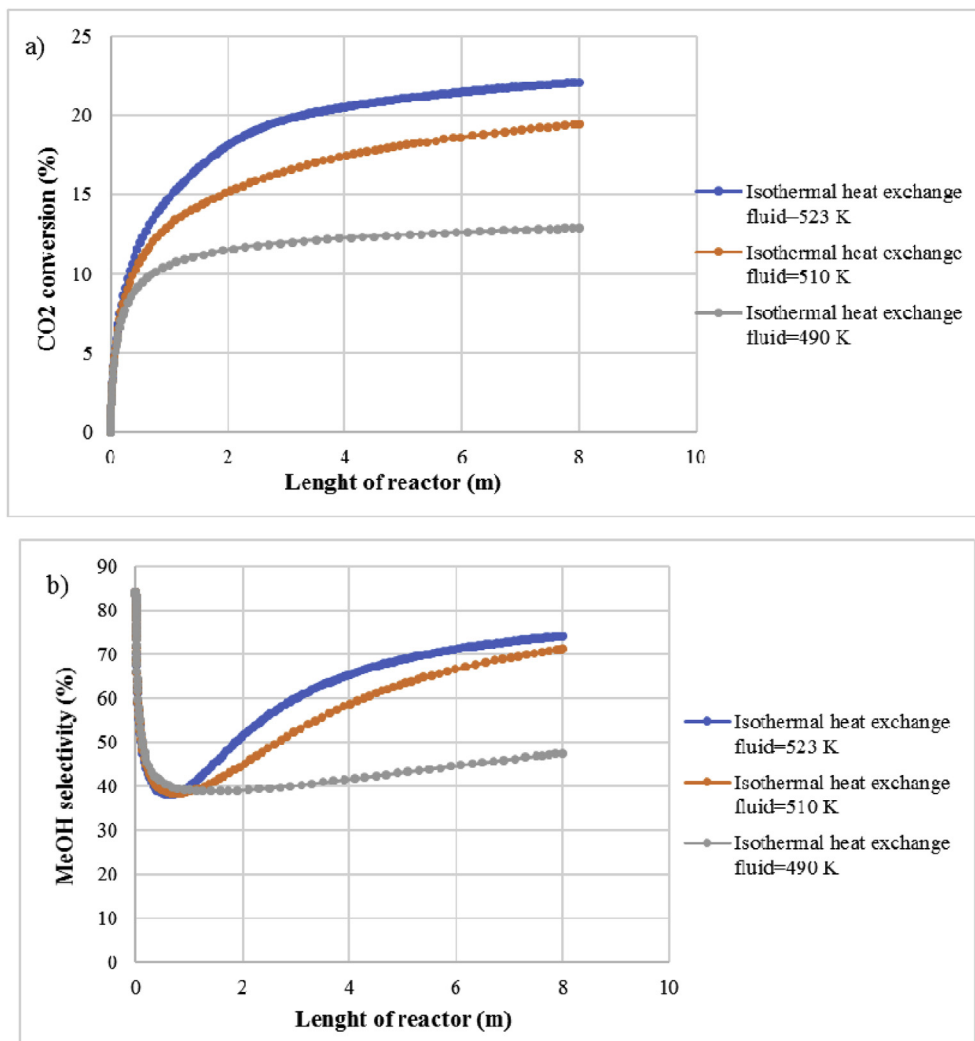
| Temperature isothermal heat exchange fluid (K)          | 523.0 | 510.0 | 490.0 |
|---------------------------------------------------------|-------|-------|-------|
| CO <sub>2</sub> flow rate at the reactor inlet (mol/s)  | 83.2  | 92.9  | 116.7 |
| CO <sub>2</sub> flow rate at the reactor outlet (mol/s) | 65.2  | 75.4  | 101.8 |
| CO <sub>2</sub> flow rate at plant inlet (mol/s)        | 25.0  | 25.0  | 25.0  |
| CO <sub>2</sub> flow rate at plant outlet (mol/s)       | 6.5   | 7.5   | 10.2  |
| MeOH production (mol/s)                                 | 13.4  | 12.1  | 6.9   |
| Temperature, reactor inlet (K)                          | 498.0 | 498.0 | 498.0 |
| Temperature, reactor outlet (K)                         | 523.6 | 510.8 | 490.6 |
| CO <sub>2</sub> conversion, in the whole process (%)    | 73.9  | 69.8  | 59.3  |
| CO <sub>2</sub> conversion, in the reactor (%)          | 21.6  | 18.8  | 12.7  |
| MeOH yield, in the whole process (%)                    | 53.6  | 48.6  | 27.4  |
| MeOH yield, in the reactor (%)                          | 16.1  | 13.1  | 5.9   |
| MeOH selectivity, in (%)                                | 74.4  | 69.5  | 46.3  |

[24], leads to different conclusions, as far as both carbon dioxide conversion and methanol selectivity are concerned.

When the value of the heat exchange coefficient at the wall is smaller, the hot spot becomes more pronounced, as expected. As a matter of facts, Fig. 8 shows that, with a reduced value of this coefficient and with the intermediate temperature level for the heat exchange fluid (510 K), a second hotspot appears at the entrance of the reactor due to high initial rate of reaction 1 and associated heat release.

In all cases examined here, temperature inside the reactor tends asymptotically to the temperature of the isothermal heat exchange fluid.

Fixing the value of global heat exchange coefficient to 1000 J/sm<sup>2</sup>K, Fig. 9a shows carbon dioxide conversion along the reactor, as predicted at different values of the isothermal heat exchange fluid temperature, while Fig. 9b shows methanol selectivity at corresponding conditions. More data are reported in Table 4.



**Fig. 10 – Additional numerical results obtained with a global heat exchange coefficient equal to 300 J/sm<sup>2</sup>K and at different values of heat exchange fluid temperature: a) Overall carbon dioxide conversion in the reactor; b) Methanol selectivity.**

**Table 5 – Calculated performance indicators for the nearly isothermal methanol reactor and the whole process with a value of global heat exchange coefficient equal to 300 J/sm<sup>2</sup>K and at different temperature of isothermal heat exchange fluid.**

| Temperature isothermal heat exchange fluid (K)          | 523.0 | 510.0 | 490.0 |
|---------------------------------------------------------|-------|-------|-------|
| CO <sub>2</sub> flow rate at the reactor inlet (mol/s)  | 83.6  | 90.7  | 115.7 |
| CO <sub>2</sub> flow rate at the reactor outlet (mol/s) | 65.2  | 73.0  | 100.8 |
| CO <sub>2</sub> flow rate at plant inlet (mol/s)        | 25.0  | 25.0  | 25.0  |
| CO <sub>2</sub> flow rate at plant outlet (mol/s)       | 6.5   | 7.3   | 10.1  |
| MeOH production (mol/s)                                 | 13.7  | 12.6  | 7.1   |
| Temperature, reactor inlet (K)                          | 498.0 | 498.0 | 498.0 |
| Temperature, reactor outlet (K)                         | 525.0 | 512.6 | 491.1 |
| CO <sub>2</sub> conversion, in the whole process (%)    | 73.9  | 70.8  | 59.7  |
| CO <sub>2</sub> conversion, in the reactor (%)          | 22.1  | 19.6  | 12.9  |
| MeOH yield, in the whole process (%)                    | 54.8  | 50.4  | 28.3  |
| MeOH yield, in the reactor (%)                          | 16.4  | 13.9  | 6.1   |
| MeOH selectivity, in (%)                                | 74.1  | 71.0  | 47.4  |

Carbon dioxide conversion at the outlet of the reactor is equal to 21.6%, 18.8% and 12.7% for isothermal heat exchange fluid temperature equal to 523 K, 510 K and 490 K, respectively. Methanol selectivity is equal to 74.4%, 69.5% and 46.3% for values of isothermal heat exchange fluid temperature equal to 523 K, 510 K and 490 K, respectively. Carbon dioxide conversion and methanol selectivity are both referred to the reactor inlet stream calculated under each process condition according to flow rate and composition of the recycle stream that is mixed with the process feeding stream (see Fig. 7). It is worth mentioning here that, as the outlet reactor temperature increases, the methanol selectivity tends to level off, because of the increasing importance of reaction 2.

On the other hand, when considering a lower value of global heat exchange, equal to 300 J/sm<sup>2</sup>K, Fig. 10a shows carbon dioxide conversion at different values of isothermal heat exchange fluid temperature, while Fig. 10b shows the corresponding methanol selectivity. Additional data are reported in Table 5. At the outlet of the reactor, carbon dioxide conversion is equal to 22.1%, 19.6%, 12.9%, for values of isothermal heat exchange fluid temperature equal to 523 K, 510 K and 490 K, respectively. Corresponding methanol selectivity at the outlet of reactor is equal to 74.1%, 71.0%, 47.4%, respectively.

These results suggest that, at fixed temperature of the heat exchange fluid, the global heat exchange coefficient has negligible influence on carbon dioxide conversion and methanol selectivity; this is probably related to the fact that, under each operating condition, the temperature reached in the reactor is quite close to the heat exchange fluid temperature, the influence of which is largely dominant. On the other hand, differences in the conversion and selectivity trends along the reactor are found according to values of the heat exchange global coefficient (Figs. 9 and 10), which in turn affect the temperature profile along the reactor, as shown in Fig. 8. It is possible to see that methanol selectivity, just after a high initial value, is decreasing in the first section of the reactor due to the temperature rise. In the second half of the reactor, methanol selectivity increases monotonically according to the temperature trend.

Overall, carbon dioxide conversion and methanol selectivity at the reactor outlet, calculated with different values of heat exchange fluid temperature, show expected trends without challenging parametric sensitivity phenomena in the different cases. For this reason, the mono-dimensional model, despite wellknown limitations in the description of local conditions inside the reactor, can be assumed as a valid computational tool in this phase of work regarding the technical feasibility of CO<sub>2</sub> hydrogenation and the choice of operating conditions.

Increasing the length of the reactor up to 12 m, i.e. the amount of catalyst in each tube and in the whole reactor, and keeping the same value of process flow rate of reactants does not affect noticeably the numerical results; the values of weight hourly space velocity (WHSV) considered in simulations allow to develop temperature, CO<sub>2</sub> conversion and MeOH selectivity profiles reaching an asymptotic trend towards the exit of the reactor. These results are also confirmed by simulations with higher values of WHSV, not reported in this work.

## Conclusions

This work was focussed on the feasibility study regarding the production of methanol from pure CO<sub>2</sub> and H<sub>2</sub> performed through the analysis of thermodynamics, plant configuration and reactor performance aspects regarding carbon dioxide hydrogenation.

Three reactor configurations were considered in the study at equilibrium conditions: once-through reactor, reactor with recycle of unconverted gases (CO, CO<sub>2</sub>, H<sub>2</sub>) after separation of methanol and water by condensation, and reactor with membrane for selective steam separation from the reaction ambient. The last two configurations were developed with the aim to overcome the thermodynamic limits present in methanol synthesis. It was found that the reactor with recycle ensures a comparatively higher efficiency. In fact, at 493 K and 55 bar, the extent of reaction 1 is equal to 38%, 69% and 58%, respectively for once through reactor, reactor with recycle and membrane reactor. As an important result, the positive effect of adding H<sub>2</sub> and CO in the feed due to the recycle is then verified.

Interesting parameters, not presented yet in previous works, as carbon dioxide conversion to carbon monoxide according to reaction 2, global carbon dioxide conversion, methanol selectivity, were also evaluated: the recycle allows to obtain the best values, except for methanol selectivity. The influence of temperature, pressure and recycle ratio highlighted at equilibrium conditions helped defining optimal operating windows at real conditions.

It was also highlighted that the use of CO<sub>2</sub> and H<sub>2</sub> in the feeding stream ensures lower  $\Delta H$  for the exothermic reaction system and therefore a reduced reactor temperature increase with reference to that obtainable with a traditional hydrogenation process, which utilizes a syngas feed stream, containing a substantial amount of carbon monoxide.

A methodology assisting the comparison and the decision making of different methanol plants based on carbon dioxide hydrogenation was proposed, by modeling the methanol

plants reported in the work of Perez-Fortes et al. [21] and Kiss et al. [18]. In both plants, a mixture of  $\text{CO}_2\text{--H}_2$  is the external feed. Results showed a good agreement between simulation results obtained in this work and literature data. The analysis of the plant illustrated in Kiss et al. [18] offered the opportunity to compare the VdB-F kinetic model (our simulations) with that proposed by Graaf et al. [31] (data reported by Kiss et al., [18]): a similar methanol throughput was predicted in our simulations; according to the VdB-F kinetic model; the recycle stream is somewhat reduced, thus allowing for a greater contact time with a consequent higher conversion of  $\text{CO}_2$  in the reaction section of the whole plant.

The simulation of a Lurgi type methanol reactor with recycle of gas reactants after separation of methanol and water by condensation was developed in MATLAB. As in the cases previously analyzed, pure  $\text{CO}_2$  and  $\text{H}_2$  were assumed in the external feeding stream. Results show very slight differences between the numerical outputs obtained with and without an axial dispersion term in the mole balance equations of the model. A sensitivity analysis varying the global heat exchange coefficient and temperature of the isothermal heat exchange fluid was also carried out, showing the dominant effect of the temperature level of the heat exchange fluid on the final temperature reached in the reactor, as well as on outlet carbon dioxide conversion and methanol selectivity.

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